Zigbee: ZigBee is an open, global, packet-based protocol that provides an easy-to-use architecture for secure, reliable, low-power wireless networks. Flow or process control equipment can be placed anywhere and communicate with the rest of the system. It can also be moved since the network doesn't care about the physical location of a sensor, pump, or valve.

Types of Testing:

1. Performance Testing

This IoT testing step covers the quickness of the correspondence network model, just as the inner calculation abilities of the inserted programming framework.

Compatibility Testing

This testing technique is executed with the conceivable blend of device adaptation, convention rendition, cell phones, and versatile OS variant.

3. End-Client Application Testing

This type of IoT testing includes the testing of all practical use instances of an IoT application which likewise incorporates client experience and convenience

Testing.

4. Security Testing

This type of IoT testing incorporates security perspectives like information assurance, gadget character confirmation, and encryption/unscrambling.

What are the IoT antenna types?

According to the use of functions, it mainly has the below 5 IoT antenna types.

(1) Radio Frequency Identification (RFID) antenna of IoT antenna types

RFID is the most important thing for the Internet of Things. One of the most commonly used IoT communication technologies.

The high-performance RFID antenna is one of the most common IoT antenna types, the RFID antenna application coverage is also the most widely used to solve the perception, reading, and identification problem of various objectives and status of the Internet of wireless sensing environment.

(2) Wearable antenna and implantable antenna of IoT antenna types

Body domain network (BAN) based on wearable and implantable communication is one of the important tip forms of the Internet of Things, which utilizes wearable antenna or implanted antennas, pick and transmit a variety of Signature information, solve the reliable communication problem between people to machine, people to people.

The wearable antenna and implantable are used in military communications, security rescue, health care, medical diagnosis, and treatment applications, etc.

(3) The Multi-physical quantity sensor antenna of IoT antenna types

Integrating various advanced functional materials on the antenna, the sensor antenna can map different physical quantities into an antenna electrical parameter, so that the antenna has a variety of physical quantities such as humidity, temperature, gas/liquid concentration, and force.

(4) Energy harvesting antenna of IoT antenna types

Energy harvesting antenna can be used as an important type of rectifier antenna, which can enrich the various radio energy in the free environment, convert it into direct current energy, and replenish the sensor node in the case of energy, to solve the terminal self-powered problem and extension of the IoT Nodes.

(5) On-chip package antenna of IoT antenna types

Combined with new materials, new physical mechanisms, and new processes, the integrated design of the entire RF front end and antenna (array) can be achieved on a piece of chip, thereby generating the conceptual design of the on-chip package antenna, which consists of the Internet of Things signal interconnection and problems with external wireless propagation.

IoT devices need to have an antenna. The smaller the space, the more the frequency band, the more complex the antenna design.

Hardware Protocols:

Low-power, short-range networks

Low-power, short range networks are well-suited for homes, offices, and other small environments. They tend to only need small batteries and are usually inexpensive to operate.

Common examples:

Bluetooth

Good for high-speed data transfer, Bluetooth sends both voice and data signals up to 10 meters.

NFC

A set of communication protocols for communication between two electronic devices over a distance of 4 cm (1/2 in) or less. NFC offers a low-speed connection with simple setup that can be used to bootstrap more-capable wireless connections.

Wi-Fi/802.11

The low cost of operating Wi-Fi makes it a standard across homes and offices. However, it may not be the right choice for all scenarios because of its limited range and 24/7 energy consumption.

Z-Wave

A mesh network using low-energy radio waves to communicate from appliance to appliance.

Zigbee

An IEEE 802.15.4-based specification for a suite of high-level communication protocols used to create personal area networks with small, low-power digital radios.

There are 6 IoT Communication Protocols/ Technology,

a. Bluetooth

An important short-range IoT communications Protocols / Technology. Bluetooth, which has become very important in computing and many consumer product markets. It is expected to be key for wearable products in particular, again connecting to the IoT albeit probably via a smartphone in many cases. The new Bluetooth Low-Energy (BLE) – or Bluetooth Smart, as it is now branded – is a significant protocol for IoT applications. Importantly, while it offers a similar range

to Bluetooth it has been designed to offer significantly reduced power consumption.

b. Zigbee

ZigBee is similar to Bluetooth and is majorly used in industrial settings. It has some significant advantages in complex systems offering low-power operation, high security, robustness and high and is well positioned to take advantage of wireless control and sensor networks in IoT applications. The latest version of ZigBee is the recently launched 3.0, which is essentially the unification of the various ZigBee wireless standards into a single standard.

c. Z-Wave

Z-Wave is a low-power RF communications IoT technology that is primarily designed for home automation for products such as lamp controllers and sensors among many other devices. A Z-Wave uses a simpler protocol than some others, which can enable faster and simpler development, but the only maker of chips is Sigma Designs compared to multiple sources for other wireless technologies such as ZigBee and others.

d. Wi-Fi

WiFi connectivity is one of the most popular IoT communication protocol, often an obvious choice for many developers, especially given the availability of WiFi within

the home environment within LANs. There is a wide existing infrastructure as well as offering fast data transfer and the ability to handle high quantities of data. Currently, the most common WiFi standard used in homes and many businesses is 802.11n, which offers range of hundreds of megabit per second, which is fine for file transfers but may be too power-consuming for many IoT applications.

e. Cellular

Any IoT application that requires operation over longer distances can take advantage of GSM/3G/4G cellular communication capabilities. While cellular is clearly capable of sending high quantities of data, especially for 4G, the cost and also power consumption will be too high for many applications. But it can be ideal for sensor-based low-bandwidth-data projects that will send very low amounts of data over the Internet.

f. NFC

NFC (Near Field Communication) is an IoT technology. It enables simple and safe communications between electronic devices, and specifically for smartphones, allowing consumers to perform transactions in which one does not have to be physically present. It helps the user to access digital content and connect electronic devices. Essentially it extends the capability of contactless card technology and enables devices to share information at a distance that is less than 4cm.

g. LoRaWAN

LoRaWAN is one of popular IoT Technology, targets wide-area network (WAN) applications. It is designed to provide low-power WANs with features specifically needed to support low-cost mobile secure communication in IoT, smart city, and industrial applications. Specifically meets requirements for low-power consumption and supports large networks with millions and millions of devices, data rates range from 0.3 kbps to 50 kbps.

IIOT:

The industrial Internet of Things (IIoT) refers to the extension and use of the internet of Things (IoT) in industrial sectors and applications. With a strong focus on machine-to-machine (M2M) communication, big data, and machine learning, IIoT enables industries and enterprises to have better efficiency and reliability in their operations. The IIoT encompasses industrial applications, including robotics, medical devices, and software-defined production processes.

Shodan:

Shodan is a search engine that focuses on internet-connected devices and services. It is designed to discover and provide information about various devices, systems, and services that are accessible over the Internet. Shodan is often referred to as the "search engine for hackers" due to its ability to identify vulnerable or misconfigured devices.

Raspberry Pi:

Raspberry Pi is defined as a minicomputer the size of a credit card that is interoperable with any input and output hardware device like a monitor, a television, a mouse, or a keyboard – effectively converting the set-up into a full-fledged PC at a low cost.

Raspberry Pi is a programmable device. It comes with all the critical features of the motherboard in an average computer but without peripherals or internal storage. To set up the Raspberry computer, you will need an SD card inserted into the provided space. The SD card should have the operating system installed and is required for the computer to boot. Raspberry computers are compatible with Linux OS. This reduces the amount of memory needed and creates an environment for diversity.

BLE:

Bluetooth Low Energy or BLE is a wireless technology used to connect devices. Bluetooth Special Interest Group (Bluetooth SIG) developed this technology focusing on industries like healthcare, security, home entertainment, etc. The BLE is Bluetooth version 4.0 and focuses on the Internet of Things (IoT) application.

BLE is a low-power wireless technology that transmits a small amount of data at lower speeds. BLE is used in low bandwidth applications, transmitting sensor data, control devices. It has low-duty data cycles and operates in a channel having a radio frequency above 40. BLE also has a quicker connection capability compared to its previous versions. Bluetooth 5 introduced in 2016 is a modification on BLE enhancing speed, range, and data capacity.

RTOS SCheduler:

RTOS scheduler—An algorithm the RTOS uses to schedule threads. The RTOS scheduler determines which thread to run on each CPU in the system at any given time.